



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Data Science Capstone



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

- Data collection from SpaceX API web scraping Wikipedia
- Data wrangling
- Exploratory data analysis with visualisations and SQL
- Interactive visualisations with Folium and Plotly Dash
- Predictive analysis with Machine Learning Classification Models

Summary of Results

- Production of csv files for use in further project work
- Wrangling of initial csv file to obtain landing outcomes for launch sites
- Exploratory data analysis of variables having an effect on landing outcomes
- Screenshots of EDA visualisations and interactive dash application
- Machine learning predictions to find the best model for landing predictions

Introduction

Project background and context

SpaceX is one of the most successful companies in commercial space flight.

The reason for this competitive edge over competitors, is due to successful landing and reuse of the first stages of rocket launches.

SpaceX is able to advertise rocket launches costing 62 million dollars, with other providers costing upwards of 165 million dollars.

Making effective predictions of whether the first stage can be reused, is essential to continue these huge savings and the competitive edge of SpaceX.

Questions to be answered

- Which variables such as booster version, launch site, payload mass, number of flights have the most effect on landing outcomes?
- Are there changes to the rate of successful landings within a range of years?
- Can we find the best machine learning classification model to predict the landing outcomes as success or failure?

Section 1

Methodology

Methodology

Executive Summary

Data collection methodology:

- Collection of data form SpaceX API
- Web scraping of Falcon 9 launch records from Wikipedia

Perform data wrangling:

- Conversion of mission outcomes to a binary value and creation of a class column

Perform exploratory data analysis (EDA) using visualisation and SQL:

- Use of graphs such bar charts, scatter plots to visualise the data
- SQL queries to find specific information on the variables in the dataset

Perform interactive visual analytics using Folium and Plotly Dash

- Creation of geospatial map showing launch site data and proximity to certain features
- Creation of a Dash interactive application with drop downs, pie charts and yearly scatter plots

Perform predictive analysis using classification models

- Data was split into training and testing sets for use in classification modeling
- The best model was determined from grid search cross validation and accuracy metrics

Data Collection

Data was collected from two sources:-

- Request for data from SpaceX API
- Scraping of Wiki page showing Falcon 9 and Heavy Launches for SpaceX

After collection of the data DataFrame were created then loaded into CSV files.

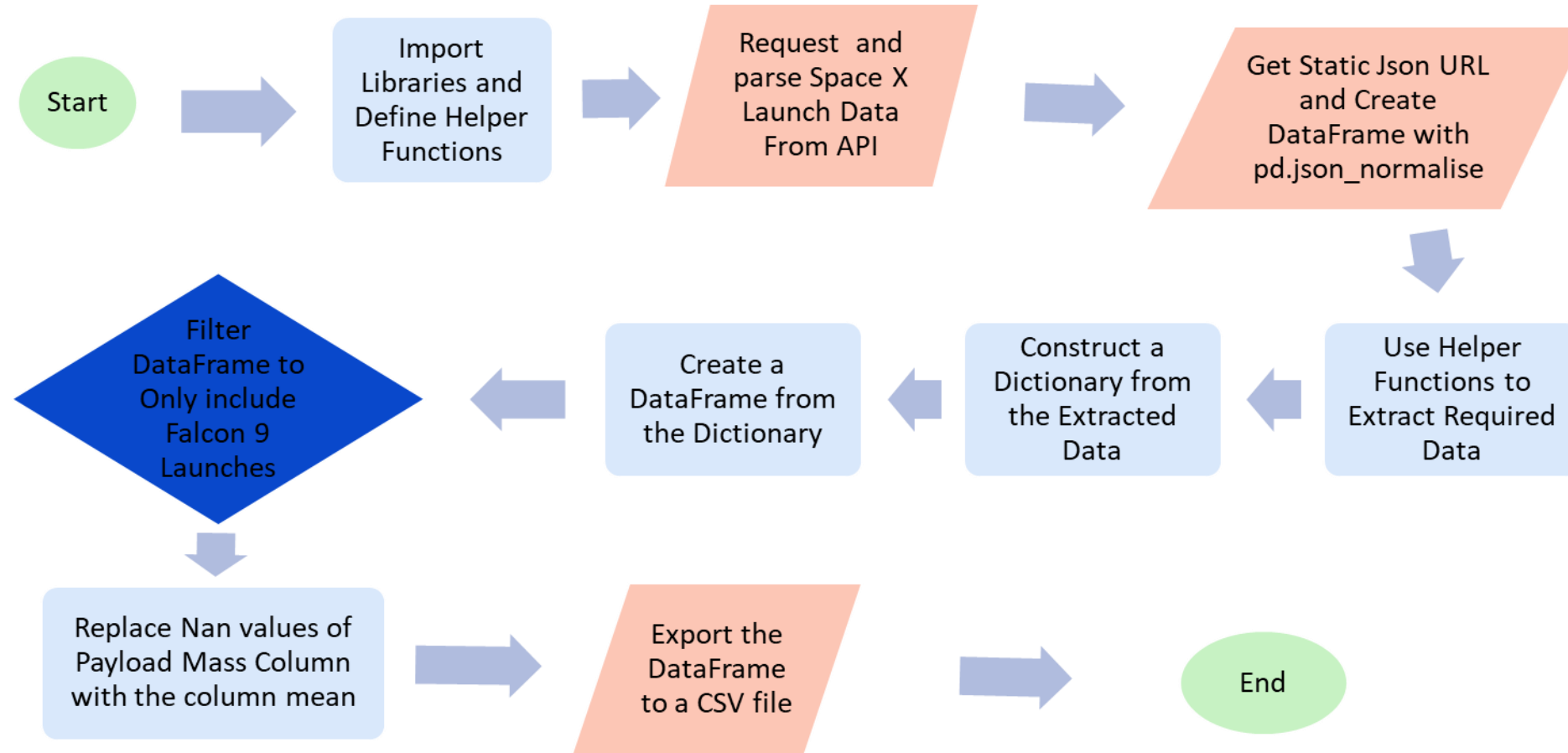
The columns obtained from SpaceX API request:-

Flight Number, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

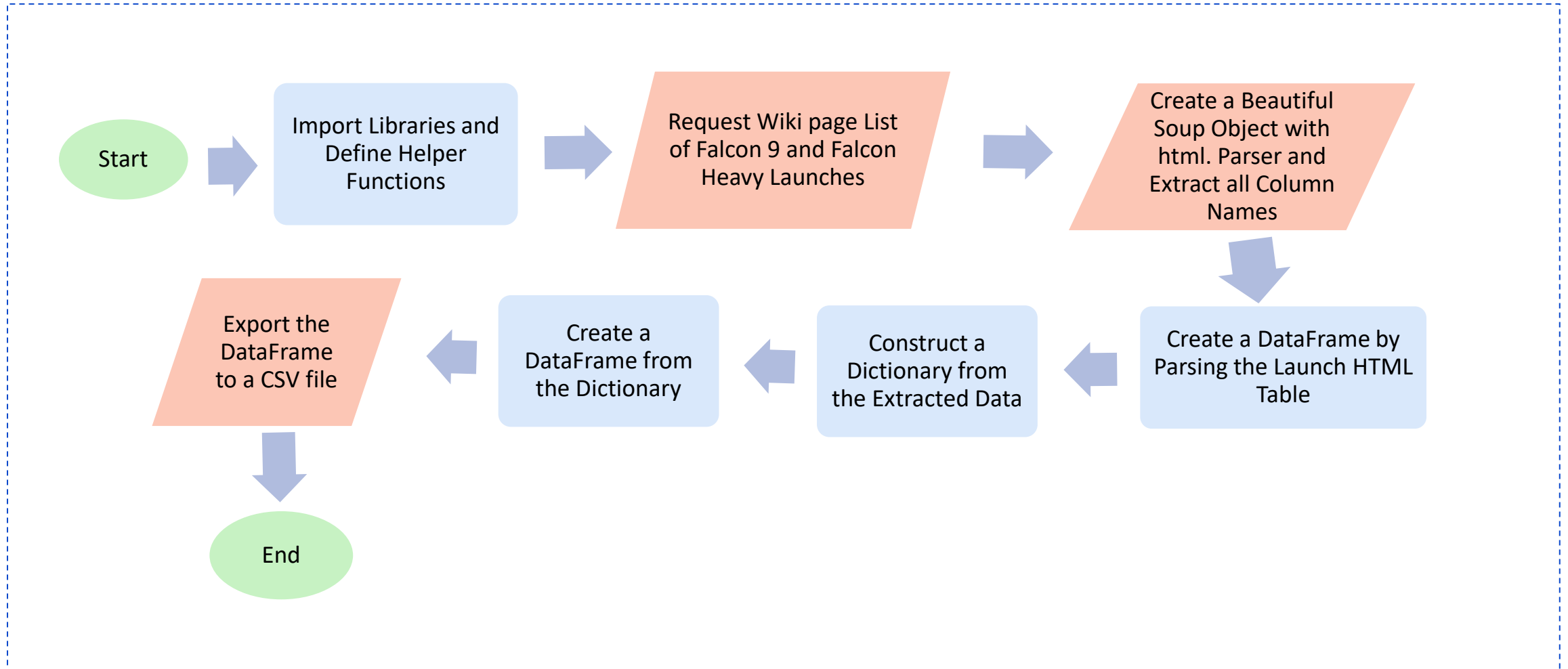
The columns obtained from Web Scraping the Wiki page:-

Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

Data Collection – SpaceX API Flow Chart



Data Collection – Web Scrapping Flowchart



Data Wrangling

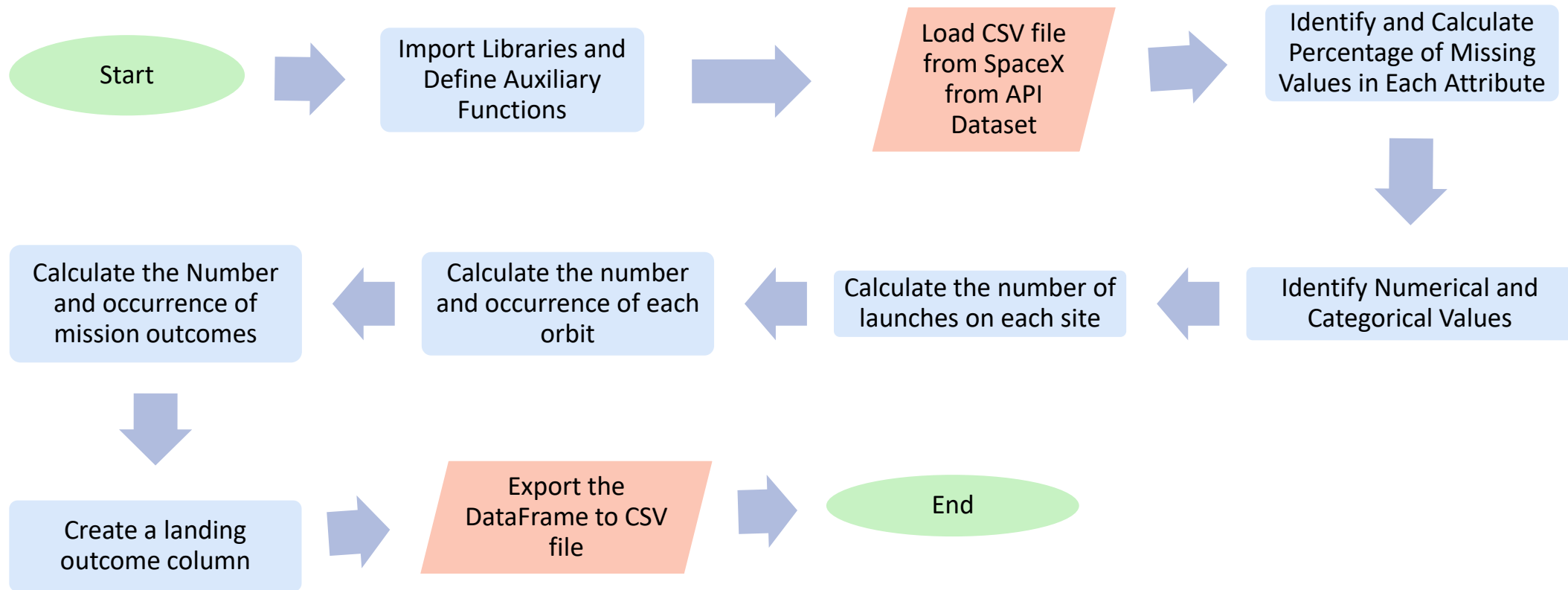
The main purpose of the data wrangling process was to extract information on launch sites, and perform some Exploratory Data Analysis (EDA) to find some patterns in the data and determine what would be the label for training supervised models.

Of particular importance was to uncover the landing outcomes for each launch site.

Outcomes were converted to a binary classification of 0 for failure and 1 for success.

A new column called “Class” was added to the DataFrame for the binary data.

Data Wrangling Flowchart



EDA with Data Visualization

Summary of the charts plotted for EDA visualisation

Scatter plots were used to visualise:-

- Flight number versus payload mass(kg), launch site and orbit
- Payload mass(kg) versus launch site and orbit

A bar chart was used to visualise:-

- The success rate for each orbit type

A line chart was used to visualise:-

- The success rate over a range of years from 2010 to 2020

Scatter plots were used to show the relationship between two variables.

A bar chart was chosen for the categorical variable orbit type to show comparisons of success rate.

A line chart was chosen for the time series as this provided the best visualisation over time.

EDA with SQL

Summary of SQL queries in EDA with SQL

- SQL queries were used to query the launch record DataFrame as follows:-
- Firstly, all columns for the DataFrame were selected
- Displayed distinct launch sites for SpaceX rockets
- Displayed 5 records where the launch site began with letters CCA
- Displayed the total payload mass carried by boosters launched by NASA (CRS)
- Displayed the average payload mass carried by booster version F9 v1.1
- Listed the date when the first successful landing in ground pad was achieved
- Listed the names of boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Listed the total number of successful and failure missions
- Listed all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function
- Listed the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015.
- Ranked the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

Build an Interactive Map with Folium

Summary of map objects such as markers, circles, lines, etc. created and added to a folium map

- A folium map was created with the centre location as NASA Johnson Space Centre at Houston, Texas.
- Circle objects were added for each launch site with an additional pop-up label showing launch site names
- Marker clusters were added to show the launch outcomes for each launch site. Green markers and red markers were allocated to successful and unsuccessful launches respectively
- A mouse position was added to the map which automatically returned latitude and longitude on scrolling through the map
- Markers were added for the closest coastlines, railways, highways and cities with distances from launch sites being shown with polylines

By adding markers, marker clusters, pop up labels and lines the map became much easier to read. Without the above there would have been no clear indication of where the launch sites are, the launch outcomes or the proximity to coastlines, railway lines etc.

[IBM-Data-Science-Capstone/Spacex - Interactive Visual Analytics with Folium \(2\).ipynb at main · Marsbars65/IBM-Data-Science-Capstone](#)

Build a Dashboard with Plotly Dash

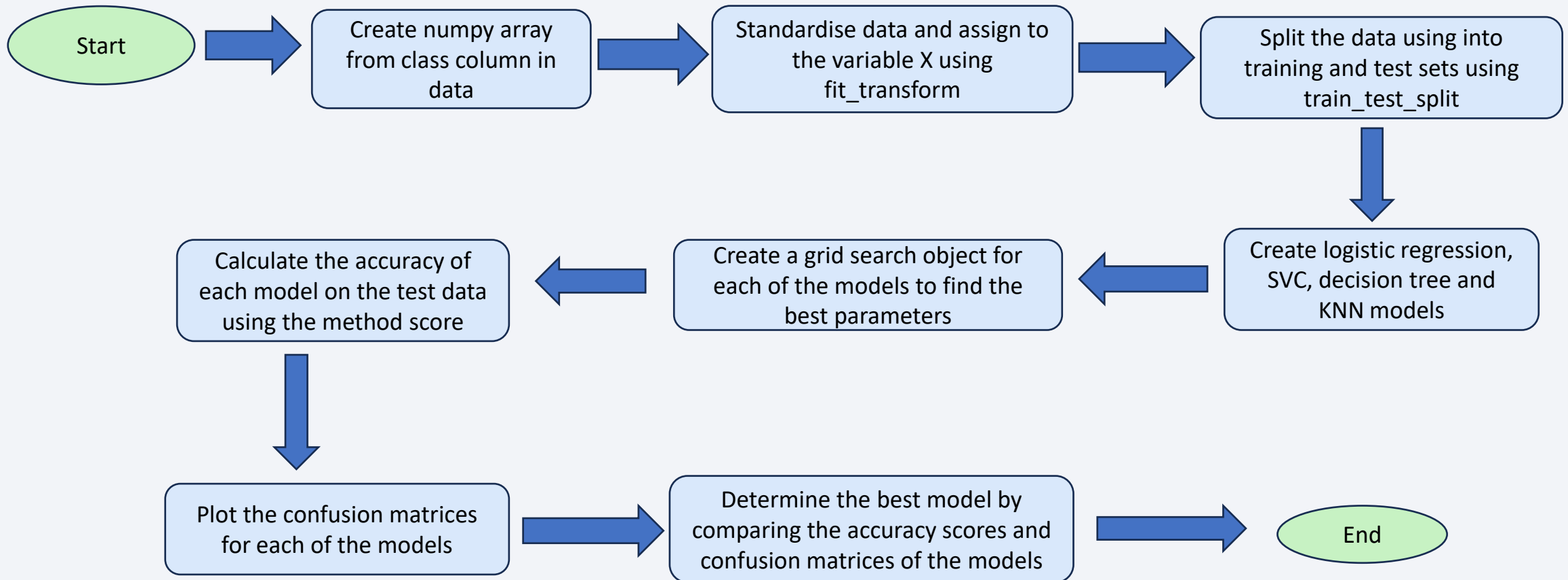
Summary of plots/graphs and interactions added to a dashboard

- A dropdown list to enable selection of all sites or specific launch sites was added
- Pie charts were added to show the success and failure rates for all sites or for specific sites
- A slider was added to enable selection of payload mass ranges
- Scatter graphs were added to show the correlation of payload mass and class of outcome for different booster versions

Adding dropdown lists, visualisations and a slider has allowed for user friendliness of the dashboard.

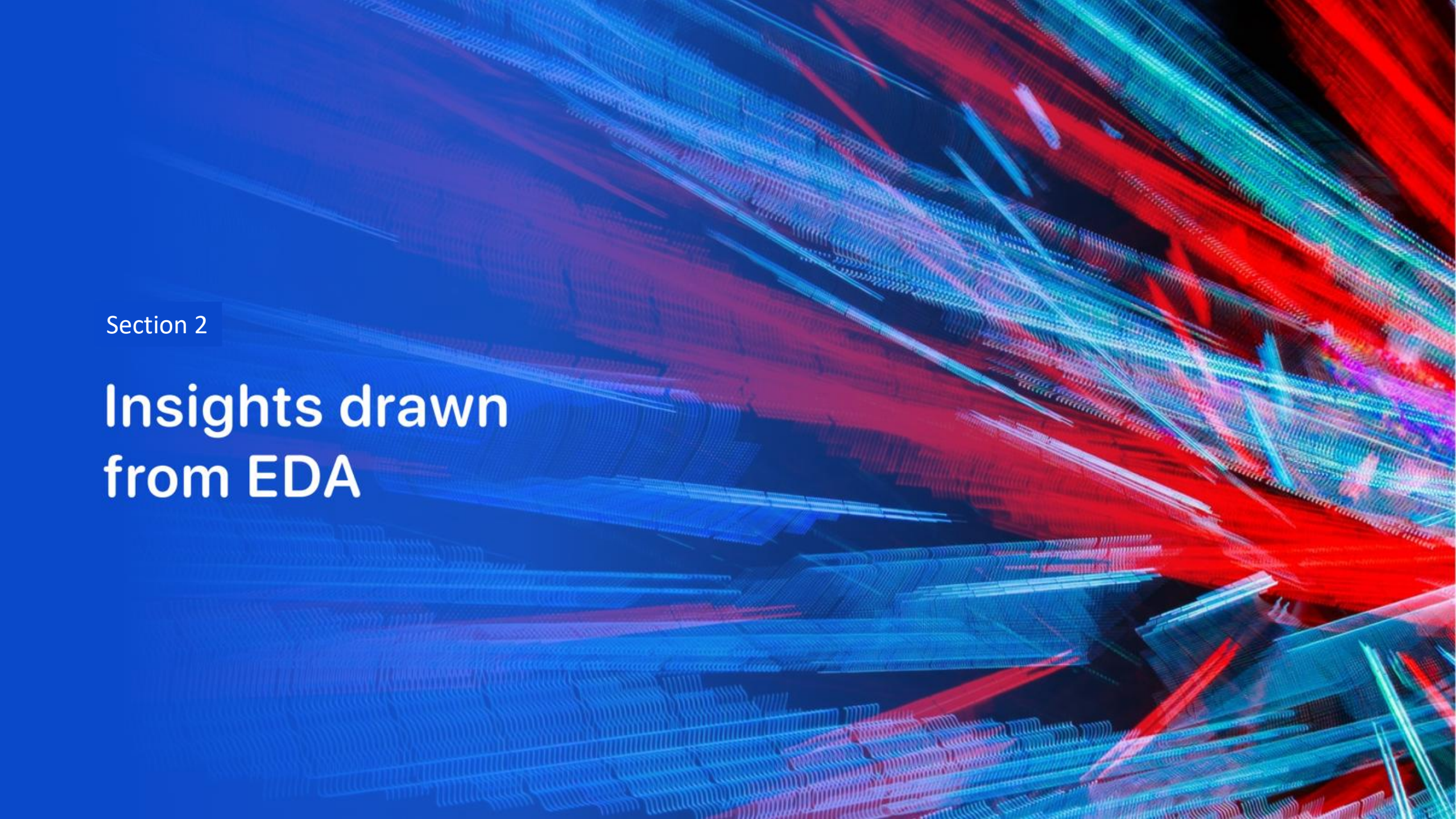
[IBM-Data-Science-Capstone/SpaceX - Launch Records Dashboard \(2\).ipynb at main · Marsbars65/IBM-Data-Science-Capstone](#)

Predictive Analysis (Classification) - Flowchart



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

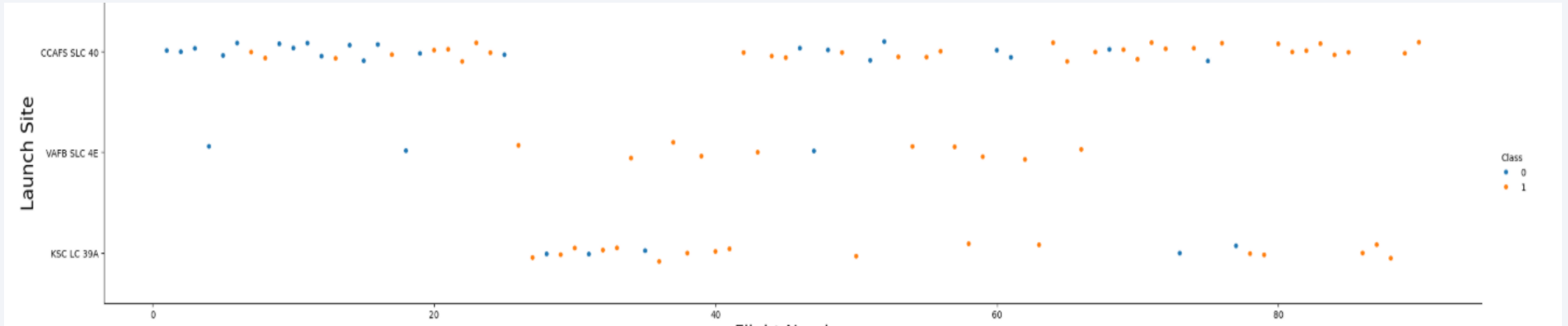
Section 2

Insights drawn from EDA

EDA with Visualisation

Flight Number vs. Launch Site

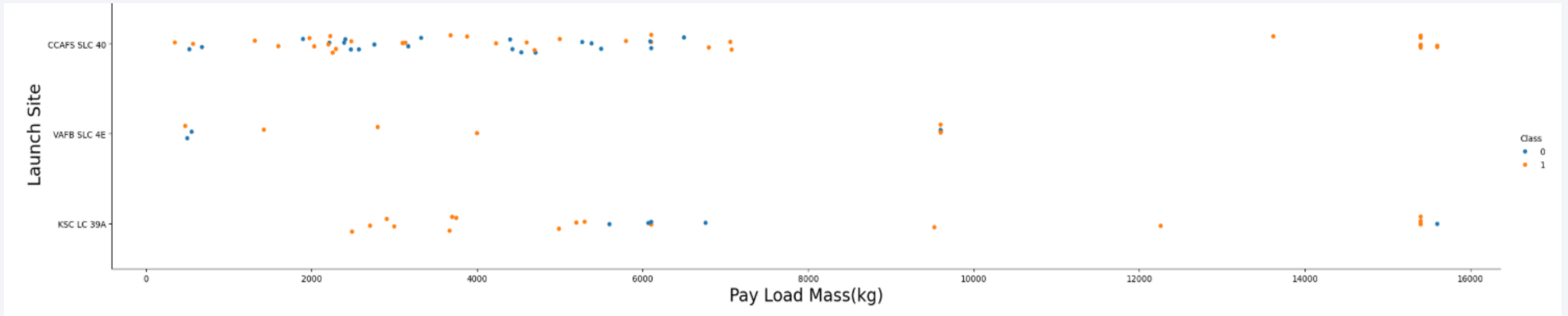
Scatter plot of Flight Number vs. Launch Site



- The majority of flights came from CCAFS SLC 40 launch site. The least number of flights was from VAFB SLC 4E.
- The number of successes appears to outnumber the number of failures for each launch site.
- The earliest flights were all unsuccessful
- The highest success rate is from the VAFB SLC 4E launch site, however there are less flights
- The latest flights have 100% success rate

Payload vs. Launch Site

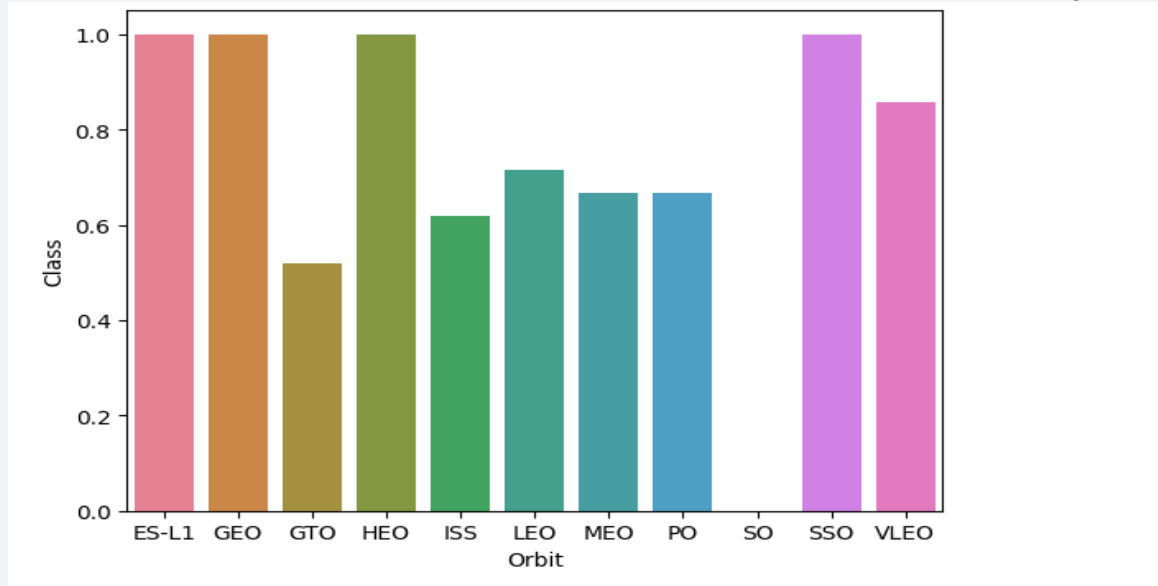
Scatter plot of Payload vs. Launch Site



- No rockets were launched with a higher payload mass than 10000 from the VAFB-SLC launch site
- At payloads higher than approximately 7000kg all launches were successful for the CCAFS_SLC launch site
- Most of the launches with a higher payload than 7000kg were successful
- The KSC LC launch site has the greatest success at lower payloads up to 6000kg

Success Rate vs. Orbit Type

Bar chart for the success rate of each orbit type



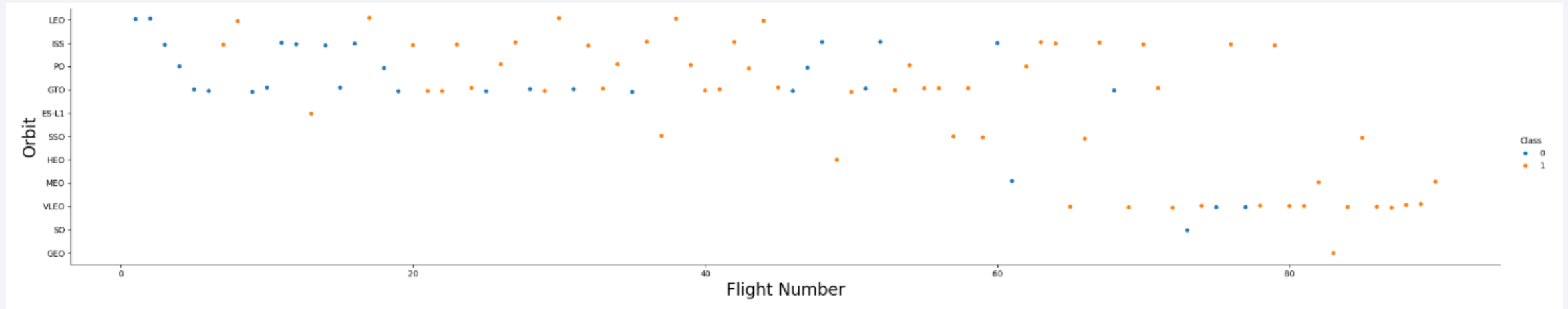
Orbits ES-LI, GEO, HEO and SSO had 100% success rate

Orbit SO had 0% success

Orbits GTO, ISS, LEO, MEO, PO and VLEO had success rates between 50% and 90%

Flight Number vs. Orbit Type

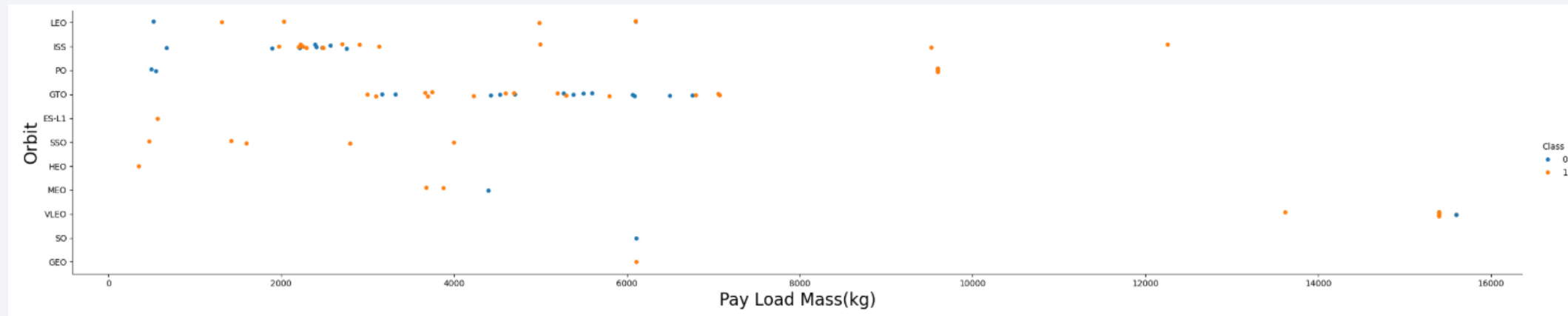
Scatter plot of Flight number vs. Orbit type



- The later flights in the LEO orbit are all successful therefore there does appear to be a correlation
- GTO orbit and number of flights appear to have no or very little correlation

Payload vs. Orbit Type

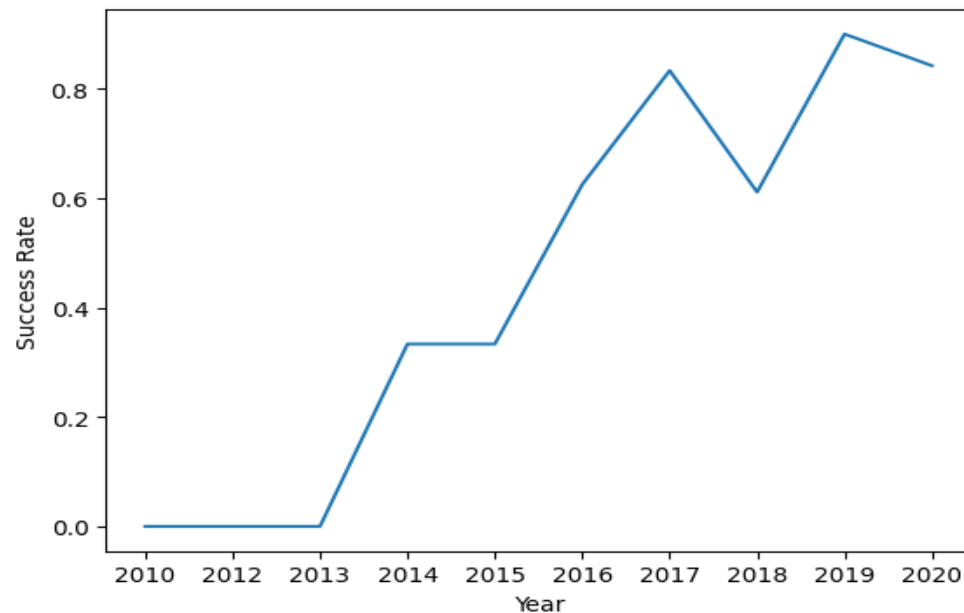
Scatter point of payload vs. orbit type



- For LEO, ISS and Polar orbits successful landing rate is greater for heavier payloads
- GTO shows a very mixed successful/failed landing rate and does not appear to be correlated with payload mass

Launch Success Yearly Trend

Line chart of yearly average success rate



- Since 2013 there has been an increase in the success rate
- Success rate stabilised from 2014 to 2015 then started to increase
- There was a sharp decrease from 2017 to 2018 and another slight decrease from 2019 to 2020

EDA with SQL

All Launch Site Names

```
%%sql
select distinct Launch_Site from spacetable
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

The SQL query selected launch sites starting with CCA from spacetable DataFrame and displayed them.

Launch Site Names Begin with 'CCA'

```
%%sql
```

Result

```
select * from spacetable where Launch_Site like 'CCA%' limit 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation

The SQL query selected launch sites starting with CCA from spacetable DataFrame and displayed them.

Total Payload Mass

```
%%sql
select sum(payload_mass_kg_) from spacetable where Customer = 'NASA (CRS)'
```

sum(payload_mass_kg_)
45596

Explanation

The query returned the total payload mass only for NASA (CRS) entries in the customer column of the spacetable DataFrame.

Average Payload Mass by F9 v1.1

```
%%sql
select avg(payload_mass_kg_) from spacetable where Booster_Version like 'F9 v1.1'
```

avg(payload_mass_kg_)
2928.4

Explanation

The query averaged the payload mass only for booster version F9 v1.1 in the booster version column of the spacetable DataFrame.

First Successful Ground Landing Date

```
%%sql
```

```
select min(date) as min_date from spacetable where "Landing_Outcome" = 'Success (ground pad)';
```

min_date
2015-12-22

Explanation

The query selected the minimum date or very first date from the landing outcome column where success was on ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
select Booster_Version from spacetable where Landing_Outcome = 'Success (drone ship)'
and payload_mass__kg_ between 4000 and 6000
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

The query selected the landing outcome success (drone ship) then limited the result to payload masses between 4000kg and 6000kg.

Total Number of Successful and Failure Mission Outcomes

```
%%sql
select Mission_Outcome, count(Mission_Outcome) from spacetable group by Mission_Outcome;
```

Mission_Outcome	count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation

The query counted the distinct mission outcomes from the spacetable DataFrame then grouped by mission outcome.

Boosters Carried Maximum Payload

```
%%sql
select Booster_Version, payload_mass_kg_ from spacetable
where payload_mass_kg_ = (select max(payload_mass_kg_) from spacetable);
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Explanation

The query used a subquery and the aggregate function select max to find the booster versions that carried the maximum payload.

2015 Launch Records

```
SELECT
CASE substr(Date, 6, 2)
  WHEN '01' THEN 'January'
  WHEN '02' THEN 'February'
  WHEN '03' THEN 'March'
  WHEN '04' THEN 'April'
  WHEN '05' THEN 'May'
  WHEN '06' THEN 'June'
  WHEN '07' THEN 'July'
  WHEN '08' THEN 'August'
  WHEN '09' THEN 'September'
  WHEN '10' THEN 'October'
  WHEN '11' THEN 'November'
  WHEN '12' THEN 'December'
END AS Month_Name,
Landing_Outcome,
Booster_Version,
Launch_Site
from spacetable
where substr(Date,0,5) = '2015'and Landing_Outcome = 'Failure (drone ship)'
```

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Explanation

The query listed the failed landing outcomes in drone ship, months, booster versions and launch sites for the year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
```

```
select Date, Booster_Version, Launch_Site, Mission_Outcome, Landing_Outcome, count(Landing_Outcome) as Landing_Counts
from spacetable where Date between '2010-06-04' and '2017-03-20'
group by Landing_Outcome
order by Landing_Counts desc;
```

Date	Booster_Version	Launch_Site	Mission_Outcome	Landing_Outcome	Landing_Counts
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Success	No attempt	10
2016-04-08	F9 FT B1021.1	CCAFS LC-40	Success	Success (drone ship)	5
2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Success	Failure (drone ship)	5
2015-12-22	F9 FT B1019	CCAFS LC-40	Success	Success (ground pad)	3
2014-04-18	F9 v1.1	CCAFS LC-40	Success	Controlled (ocean)	3
2013-09-29	F9 v1.1 B1003	VAFB SLC-4E	Success	Uncontrolled (ocean)	2
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Success	Failure (parachute)	2
2015-06-28	F9 v1.1 B1018	CCAFS LC-40	Failure (in flight)	Precluded (drone ship)	1

Explanation

Landing outcomes ranked by landing counts between 4th of June 2010 and 20th of March 2017.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in the lower right quadrant of the image, following the curve of the Earth. The upper left quadrant shows the dark blue sky with a few stars. The horizon line of the Earth is visible, separating the dark sky from the illuminated surface.

Section 3

Launch Sites Proximities Analysis

All launch sites' location markers on a global map

Explanation

- All launch sites are close to a coastline. This is for safety as rockets can be launched seaward to ensure falling debris does not land near any populations of people
- All launch sites are near the equator which takes advantage of the Earth's rotational speed, thereby providing extra velocity boost which reduces fuel consumption and allows more efficient access to various orbital inclinations (ref: Microsoft Copilot Search)



Colour-labeled launch outcomes on the map

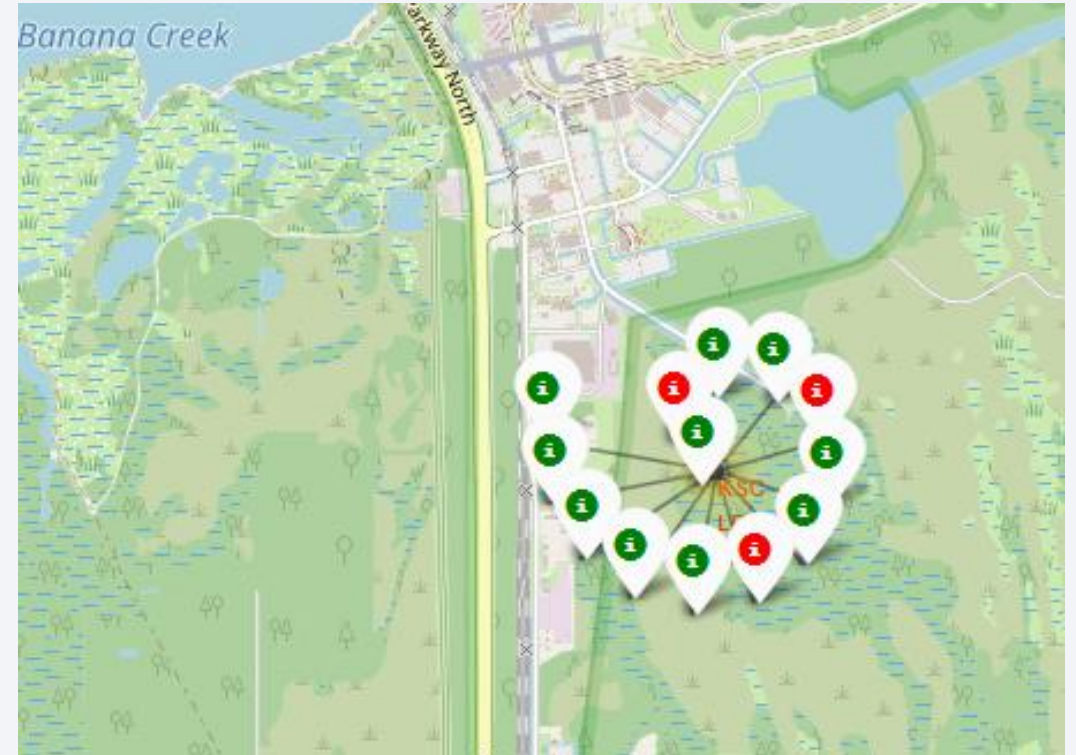
Explanation

- Colour markers allow for easy identification of launch sites with high success rates as follows:-

Green Marker = Successful Launch

Red Marker = Failed Launch

- The KSC LC-39A has a high success rate as shown in the screenshot opposite with 10 out of 13 launches being successful



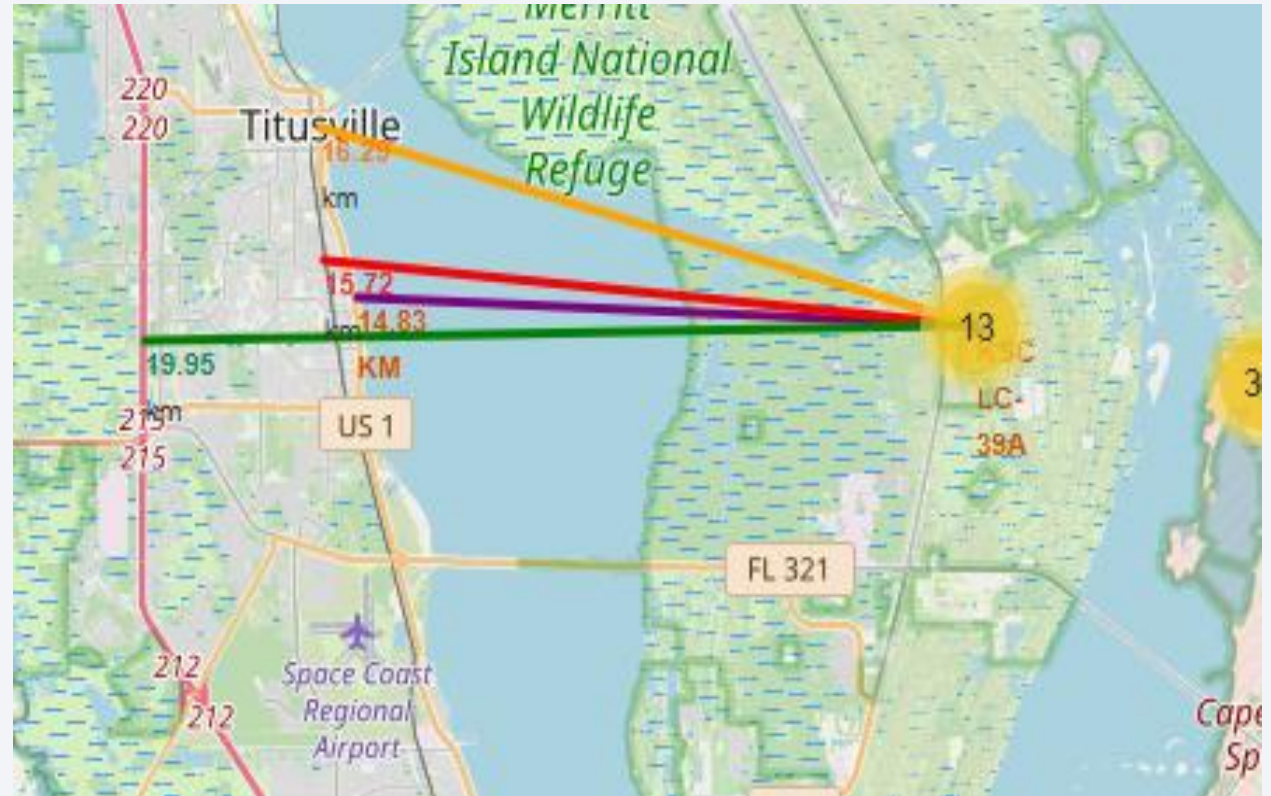
Distances to proximities from KSC-LC-39A Launch Site

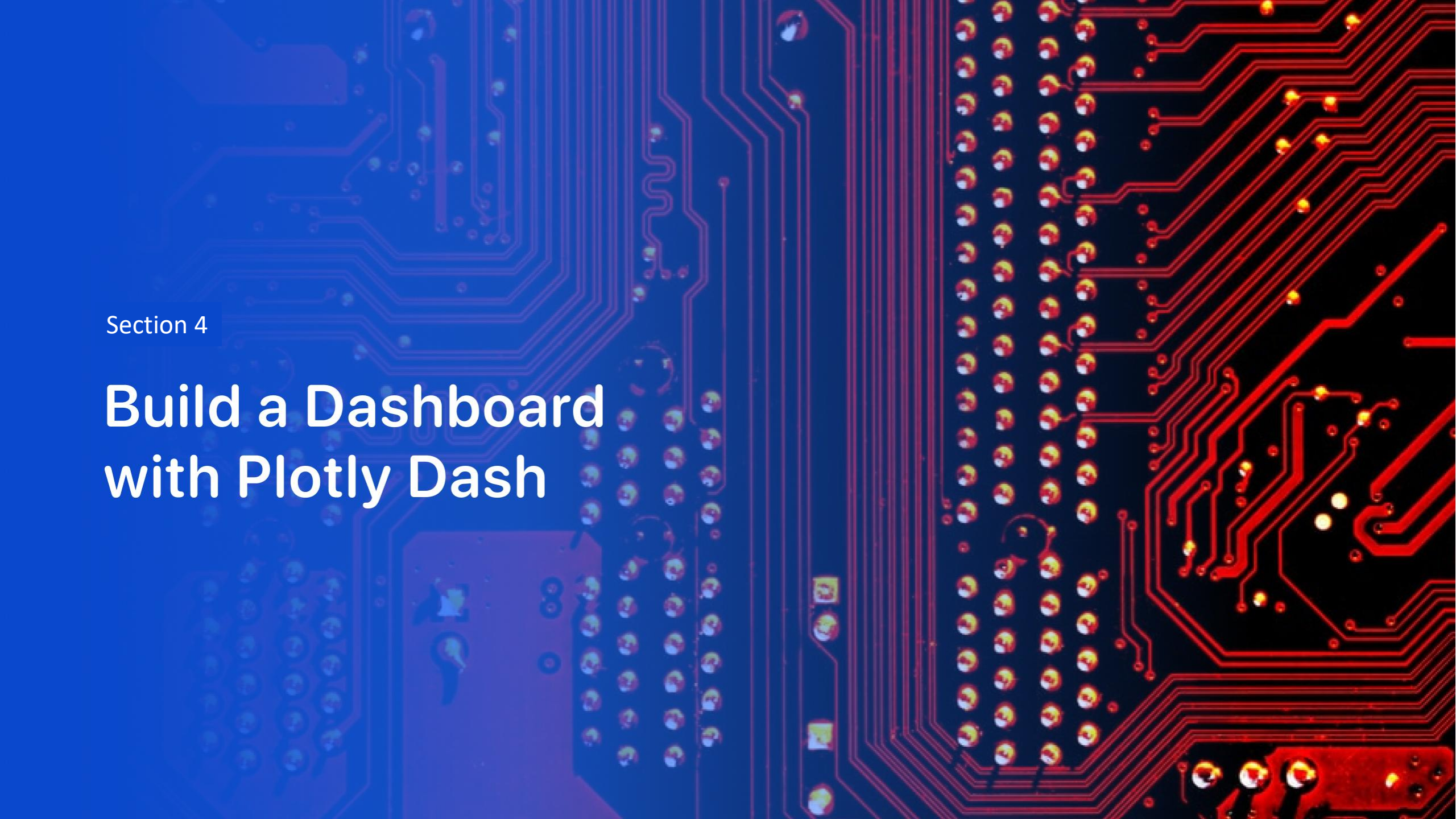
Explanation

Distances from the KSC-LC-39A launch site to proximities are:-

Railway line	15.72km
Coastline	14.83km
Highway	19.95km
City(Titusville)	16.25km

The above are quite close to the launch site. The distance to the nearest city is of particular note. This could be potentially dangerous if there was falling debris from a rocket launch.

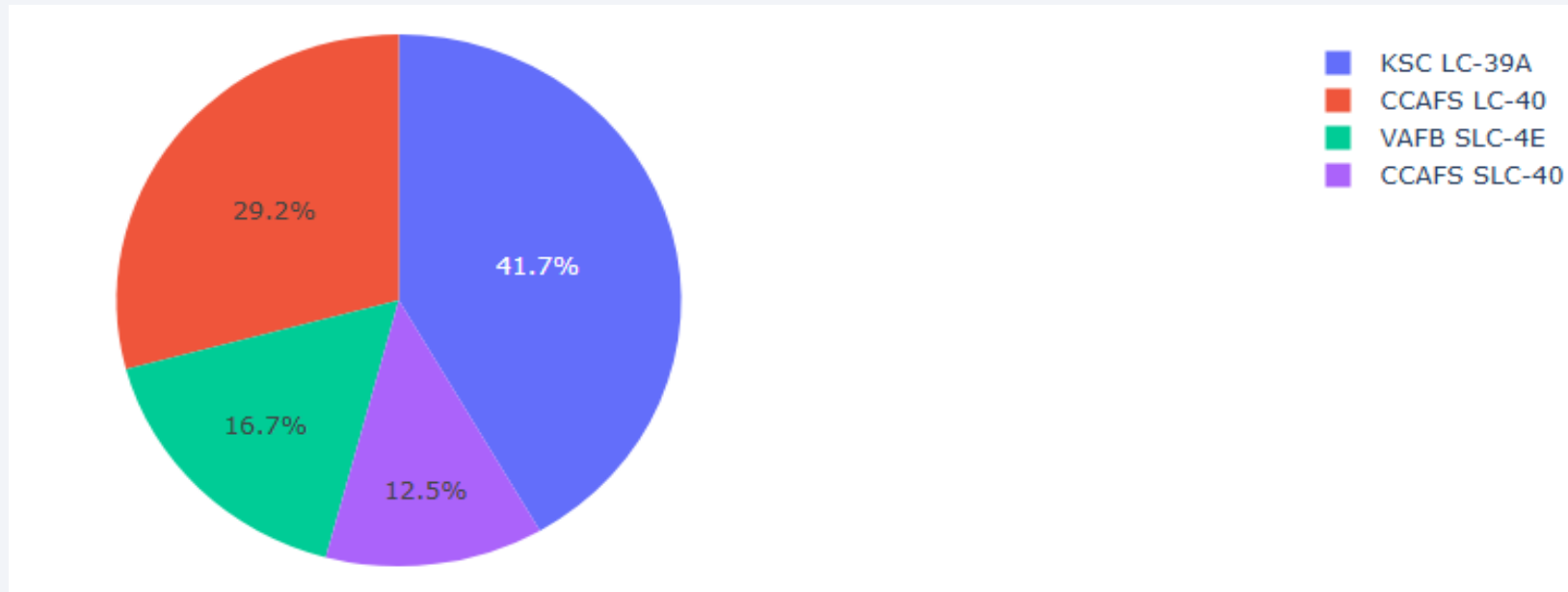




Section 4

Build a Dashboard with Plotly Dash

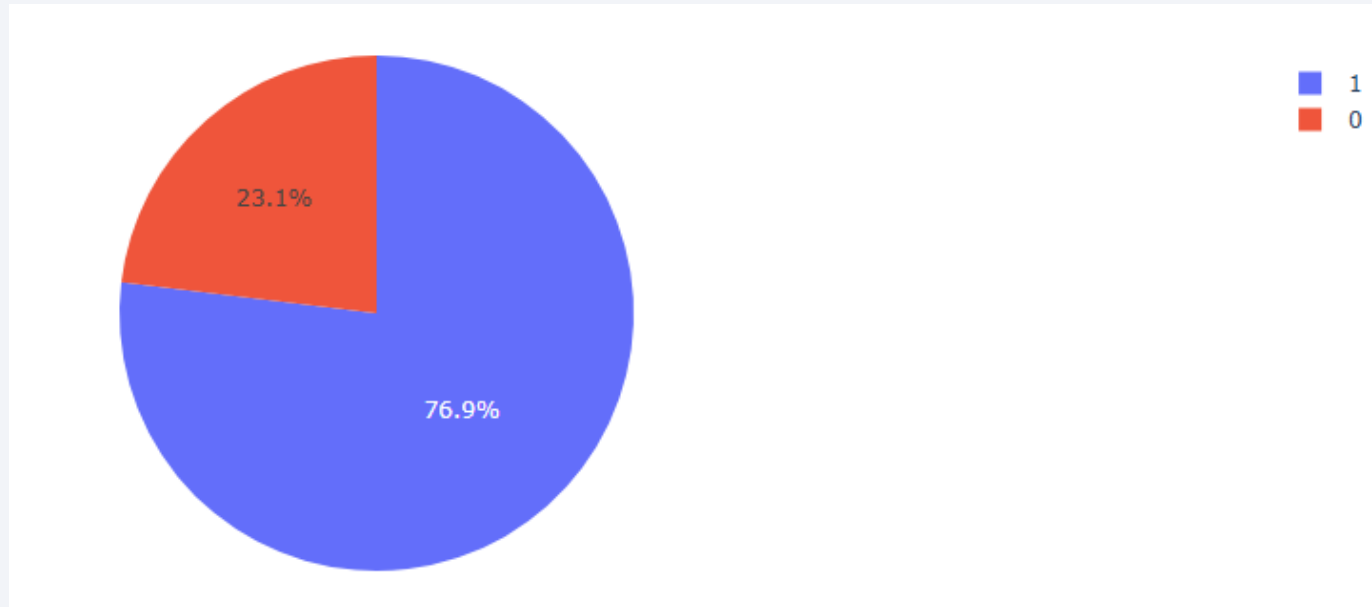
Total Successful Launches by Site



Explanation

From the pie chart and the key opposite it can be seen that the launch site with the highest success rate is the KSC-LC-39A launch site with a rate of 41.7%. Only 8.3% less than half of all launches.

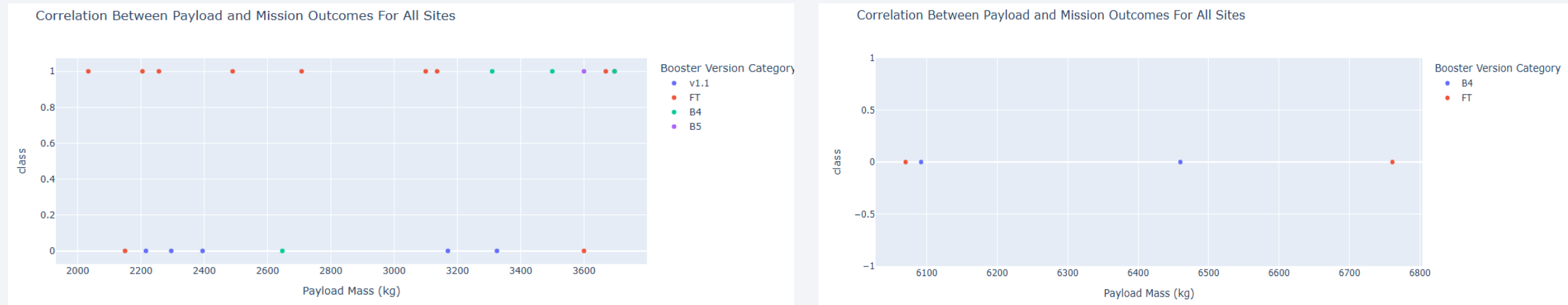
Total Success/Failure Rate for Launch Site KSC-LC-39A



Explanation

The success rate of rocket launches at KSC-LC-39A launch site is 76.9% as opposed to a failure rate of 23.1%. Also, the folium map showed that 10 out of 13 landings were successful.

Payload Mass vs. Launch Outcome for all sites



Explanation

Payload masses between 2000kg and 4000kg have a greater success rate.

Heavier payloads between 6000kg and 8000kg had no successful landings as can be seen from the plot on the right.

Booster version FT has the highest success rate as can be seen from the plot on the left.

There is one out of one successful rocket launch for the B5 rocket, however this would not provide a fair comparison for the other booster versions.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

The built model accuracy for all built classification models is shown below:-

Model	Precision	Recall	F1 Score	Accuracy
Logistic Regression	0.80	1.00	0.89	0.83
Support Vector	0.80	1.00	0.89	0.83
Decision Tree	0.92	0.92	0.92	0.89
K Nearest Neighbour	0.80	1.00	0.89	0.83

The model with the highest classification accuracy is the Decision Tree model with an accuracy of 0.89 versus an accuracy of 0.83 for the other models.

For three of the models the classification report metrics are the same which is due to the small sample size.

**The classification reports for the models are shown in Appendix A*

Confusion Matrix

The confusion matrix for the Decision Tree model is shown opposite.

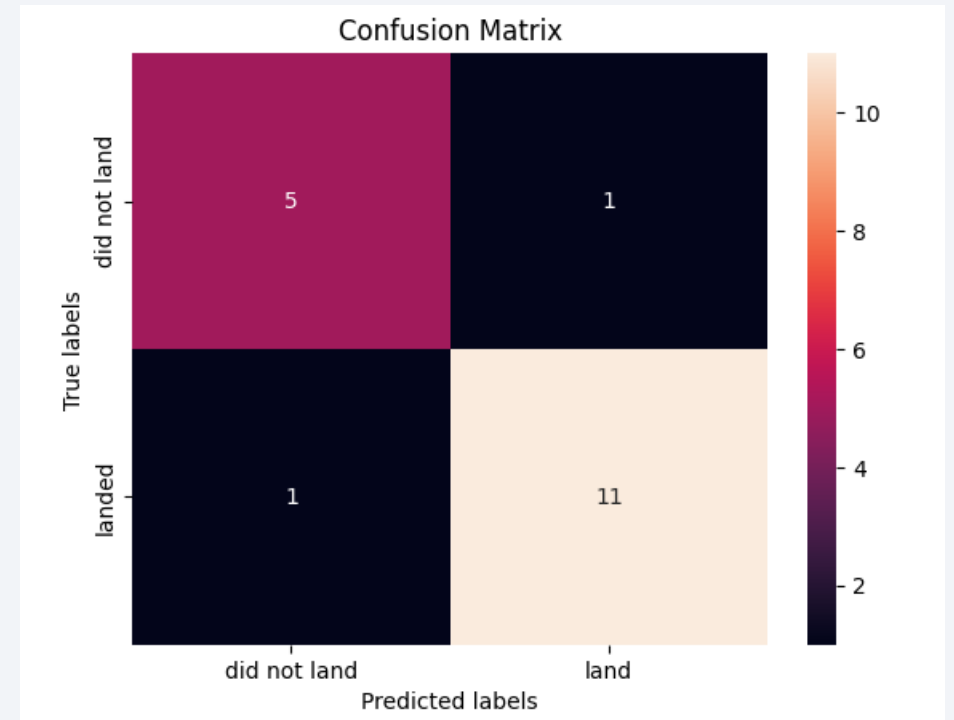
Explanation

The confusion matrix represents the following:-

True Positive	11
True Negative	5
False Positive	1
False Negative	1

16 out of 18 landing outcomes were predicted accurately.

**See Appendix B for confusion matrix of other models*



Conclusions

- The KSC-LC-39A has the highest success rate of landing outcomes compared to other launch sites.
- There is an upwards trend in yearly success rates with a dip between 2017 to 2018 perhaps due to significant operational challenges(ref Microsoft co-pilot). There was another dip between 2019 to 2020 perhaps due to covid-19.
- Lower payload mass results in a higher rate of launch successes than high payload mass.
- Launch sites are in close proximity to a railway line, highway, coastline or city.
- All launch sites are close to the equator to take advantage of the higher rotational speed of the Earth.
- The best model for predicting landing outcomes is the Decision Tree Model.
- Orbits ES-LI, GEO, HEO and SSO had 100% success rate.

Appendix

Appendix A

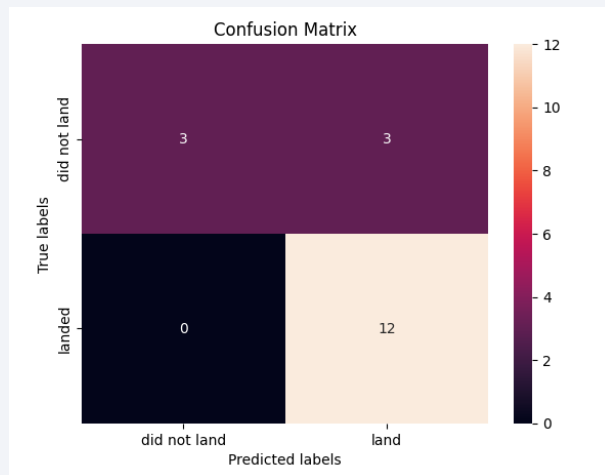
Decision Tree Model

	precision	recall	f1-score	support
0	0.83	0.83	0.83	6
1	0.92	0.92	0.92	12
accuracy			0.89	18
macro avg	0.88	0.88	0.88	18
weighted avg	0.89	0.89	0.89	18

SVC, KNN and Logistic Regression Models

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Appendix B



Thank you!

